

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain in detail various Arithmetic operations on intervals and numbers. (CO2)

Q.24 Explain Evidence and Possibility Theory in detail. (CO3)

Q.25 Write short note on the following.

a) Extension Principle of fuzzy sets (CO2)

b) Crisp relations (CO3)

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Roll No.

6th Sem / Artificial Intelligence & Machine learning

Subject : Fuzzy Logic and Applications

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The range of Fuzzy set is : (CO1)

- a) 0 to 1 b) -1 to 1
- c) 0 to 100 d) -100 to 100

Q.2 What is the result of multiplying two fuzzy numbers? (CO2)

- a) A crisp number
- b) A fuzzy number
- c) A random number
- d) An uncertain number

Q.3 What is the basic concept of Fuzzy Logic? (CO1)

- a) Binary logic b) Multivalued logic
- c) Probabilistic logic d) Deterministic logic

Q.4 The Arithmetic operation of Fuzzy Numbers is: (CO2)

- a) Addition b) Multiplication
- c) Division d) All

Q.5 Compatibility relation concept is based on: (CO3)

- a) Equivalence b) Order
- c) Compatibility d) All

Q.6 The application of Fuzzy Logic is: (CO4)

- a) Automotive systems b) Defence
- c) Securities d) All

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 A fuzzy arithmetic is arithmetic operations on _____ numbers. (fuzzy/crisp) (CO2)

Q.8 Crisp sets have the range of _____. ($\{0,1\}$ / (0 to 1)) (CO1)

Q.9 Union is a membership function in fuzzy logic. (True/False) (CO1)

Q.10 A fuzzy qualifier is a linguistic term that modifies a _____ set. (fuzzy/ Crisp) (CO3)

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Q.11 The purpose of using fuzzy arithmetic is to perform exact calculations. (True/False) (CO2)

Q.12 Fuzzy Logic is used in pattern recognition. (True/False) (CO4)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain general aggregation fuzzy operations. (CO2)

Q.14 Write the properties of α -cuts. (CO2)

Q.15 Write the applications of Fuzzy Logic in electronic devices. (CO4)

Q.16 What are Fuzzy Relation Equations? Give one example. (CO3)

Q.17 Explain any two Membership Functions. (CO1)

Q.18 Explain Binary Fuzzy Relations. (CO3)

Q.19 What are Linguistic Hedges? Give one example. (CO3)

Q.20 Differentiate Crisp sets and fuzzy sets. (CO1)

Q.21 How fuzzy Logic is helpful in business? (CO4)

Q.22 What are Linguistic variables? Give two examples. (CO2)

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